

CRISP DM Audit

CHATGPT

1. Class ranking

Ranking criterion: class-level F1, because it balances precision and recall.

Rank	Class	Precision	Recall	F1	Key failure characteristic
1	Forest	0.9874	0.9916	0.9895	Only 4 of 474 images misclassified
2	Sea	0.9072	0.9588	0.9323	High recall, but lower precision from glacier/mountain images predicted as sea
3	Street	0.9430	0.8922	0.9169	49 street images predicted as buildings
4	Buildings	0.8849	0.9497	0.9161	High recall but lower precision, partly because street images enter this class
5	Mountain	0.8608	0.8248	0.8424	69 mountain images predicted as glacier
6	Glacier	0.8423	0.8210	0.8315	63 glacier images predicted as mountain and 30 as sea

Observed evidence: Forest has the highest F1 and glacier the lowest. Street narrowly exceeds buildings in F1, although buildings has higher recall.

Interpretation: Forest is the strongest class on this test set. Glacier and mountain are the weakest because of substantial reciprocal confusion. Street and buildings have similar F1 scores but different failure profiles: street has stronger precision, while buildings has stronger recall.

Proposed validation experiment: Repeat the class ranking across multiple seeds and bootstrap the test predictions to obtain confidence intervals for each F1. Without uncertainty estimates, small differences—especially street versus buildings—should not be treated as a stable ordering.

2. Largest directional confusions

Rank	Actual class → predicted class	Count	Percentage of actual class
1	Mountain → glacier	69	13.1%
2	Glacier → mountain	63	11.4%
3	Street → buildings	49	9.8%
4	Glacier → sea	30	5.4%
5	Buildings → street	20	4.6%
6	Mountain → sea	16	3.0%
7	Sea → glacier	14	2.7%

Observed evidence: Mountain–glacier confusion is strongly bidirectional: 132 images cross between the two classes. Street–

buildings confusion is asymmetric: 49 street images are classified as buildings, while only 20 building images are classified as street.

Interpretation: The mountain–glacier boundary is the model’s dominant class-boundary problem. The street–buildings boundary appears biased toward predicting buildings when an urban image contains both road and architectural content.

Proposed validation experiment: Export sample-level predictions and manually categorize every error in these three directions by visible attributes—snow/ice, exposed rock, water, roadway area, façade area, skyline, and mixed-scene composition. Test whether these attributes occur significantly more often in errors than in correctly classified images.

3. Glacier–mountain confusion

Shared terrain and texture

Observed evidence: The matrix contains 69 mountain→glacier and 63 glacier→mountain errors. Glacier and mountain have the two lowest recalls, 0.8210 and 0.8248.

The Grad-CAM gallery includes:

- A mountain predicted as glacier at 98.5% confidence
- A glacier predicted as mountain at 98.9% confidence

Their heatmaps overlap prominent terrain regions.

Interpretation: Both classes can contain steep slopes, ridgelines, exposed rock, snow-like light areas, and similar blue-gray-brown textures. At 150×150 resolution, fine distinctions between snow, ice, scree, and exposed rock may be weakened.

This explanation is plausible but not proven. The artifacts do not show whether resolution caused these errors.

Proposed validation experiment: Re-evaluate the same trained design at multiple input resolutions, such as 150×150, 224×224, and a higher resolution, using identical splits and repeated seeds. Test specifically whether glacier→mountain errors decrease as fine surface detail is preserved.

Mixed scenes and contextual cues

Observed evidence: The error gallery shows mountainous scenes containing combinations of slopes, snow-like material, sky, and water. One glacier→mountain example receives 98.9% confidence, and one mountain→glacier example receives 98.5%.

Interpretation: The classifier may be relying on broad scene composition or texture rather than a precise distinction between glacial ice and mountain terrain. A glacier image dominated by rock may resemble the mountain class, while a snowy mountain may resemble the glacier class.

Proposed validation experiment: Create controlled crops or masks for the same images:

- Terrain-only
- Sky removed
- Water removed
- Snow/ice region masked
- Rock region masked

Compare probability changes. If masking snow/ice sharply reduces glacier probability, that would support—but still not conclusively prove—use of that cue.

Sea-related glacier errors

Observed evidence: Thirty glacier images are predicted as sea. Two glacier→sea examples appear in the Grad-CAM gallery at 99.6% and 98.6% confidence. Their heatmaps emphasize broad horizontal water/sky or shoreline-like regions.

Interpretation: Water, blue color, horizon structure, and reflective surfaces may dominate the prediction when the glacial component is small or visually weak. Some glacier images may also contain lakes or coastal-looking regions.

Proposed validation experiment: Compare errors on glacier images with and without visible water. using a blinded human

annotation. Estimate glacier recall separately for the two subgroups.

4. Street–buildings confusion

Urban co-occurrence

Observed evidence: Forty-nine of 501 street images are predicted as buildings, compared with 20 of 437 building images predicted as street. The supplied street→buildings example is wrong at 99.8% confidence.

Its Grad-CAM activation overlaps a prominent building/tower region rather than being restricted to the road surface.

Interpretation: Streets and buildings frequently coexist. If architecture occupies more of the frame or is more visually distinctive than the roadway, the model may favor buildings even when the folder label is street.

The directional asymmetry suggests that architectural evidence may dominate road evidence in some mixed urban scenes. This remains an interpretation, not a proven internal rule.

Proposed validation experiment: Measure façade area and visible-road area using manual masks or a segmentation model.

Compare these ratios across:

- Correctly classified streets
- Street→buildings errors
- Correctly classified buildings
- Buildings→street errors

Test whether street→buildings errors contain systematically more façade area or less visible roadway.

Landmark and architectural dominance

Observed evidence: The 99.8%-confidence street→buildings example prominently contains a large clock tower and surrounding architecture. The Grad-CAM hotspot overlaps the architectural structure.

Interpretation: A distinctive landmark may be a stronger feature than the lower-level street region. The prediction may therefore be visually understandable even if it disagrees with the dataset label.

Proposed validation experiment: Mask the tower/building region and recompute predictions. Separately mask the road and pedestrian region. A larger probability change after architectural masking would support the landmark-dominance hypothesis.

5. Model error versus possible label ambiguity

Clear model error

Observed evidence: A prediction disagrees with the supplied folder label. For evaluation, every such disagreement is counted as an error.

Interpretation: It is defensible to call it a model error relative to the dataset ground truth. It is not always defensible to call it an unequivocal semantic error, because the images may contain multiple valid scene concepts.

Proposed validation experiment: Have several blinded reviewers assign:

- One dominant label
- All applicable labels
- An ambiguity score

A disagreement supported by strong reviewer consensus is more likely to be a genuine model error.

Possible glacier–mountain ambiguity

Observed evidence: The gallery includes glacier and mountain images containing overlapping terrain characteristics. The confusion

...between the glacier, mountain, glacier, and mountain images containing overlapping terrain and vegetation. The confusion is reciprocal rather than entirely one-directional.

Interpretation: Some errors may result from an under-specified boundary between glacier and mountain. For example, an image labeled glacier may be visually dominated by rock, while a mountain image may contain extensive snow or ice.

Reciprocal confusion is consistent with ambiguity, but it does not prove annotation ambiguity.

Proposed validation experiment: Ask domain-informed reviewers to relabel the 132 glacier→mountain errors without seeing the original labels or model predictions. Report consensus and inter-rater agreement.

Possible street→buildings ambiguity

Observed evidence: The street→buildings example contains both a street-level scene and prominent architecture.

Interpretation: “Street” and “buildings” may describe different valid components of the same image. The dataset’s single-label structure forces one answer even when multiple concepts are present.

Proposed validation experiment: Compare single-label and multilabel annotation for urban error cases. If reviewers consistently select both street and buildings, part of the measured confusion reflects task formulation rather than purely defective recognition.

6. What Grad-CAM supports

Observed evidence: In the six selected confident errors, Grad-CAM highlights spatial regions associated with architecture, water/horizons, slopes, and snow- or rock-like terrain.

Interpretation: Grad-CAM supports the limited statement that the model’s predicted-class score is locally sensitive to feature-map activations spatially associated with those regions. It helps generate hypotheses about possible cues.

Proposed validation experiment: Perform Grad-CAM sanity checks by progressively randomizing classifier and backbone weights. A meaningful explanation method should change substantially as learned parameters are destroyed.

7. What Grad-CAM cannot prove

Observed evidence: Only six deliberately selected, highly confident errors are visualized. No random errors, correct cases, alternative explanation methods, or perturbation tests are supplied.

Interpretation: The figure cannot prove:

- That highlighted pixels caused the prediction
- That non-highlighted regions were irrelevant
- That the model recognized concepts such as “ice,” “road,” or “building”
- That the explanations are faithful or stable
- That the same behavior occurs across the full test set
- That errors arise from bias rather than label ambiguity
- That the model is globally miscalibrated

The heatmaps are qualitative hypotheses, not causal explanations.

Proposed validation experiment: Compare Grad-CAM with occlusion sensitivity and an attribution method such as Integrated Gradients. Quantitatively test whether removing highly attributed regions reduces the predicted-class score more than removing equally sized random regions.

Three priority experiments

1. Blinded ambiguity audit

Independently relabel glacier→mountain and street→buildings errors using multiple reviewers, multilabel choices, confidence ratings, and inter-rater agreement

ratings, and inter-rater agreement.

2. **Counterfactual region intervention**

Mask or crop snow/ice, rock, water, road, and architecture regions. Measure changes in class probabilities and predictions against random-region controls.

3. **Controlled subgroup challenge set**

Evaluate separately on:

- Pure glacier versus glacier-with-rock/water
- Pure mountain versus snowy mountain
- Road-dominant streets versus architecture-dominant streets
- Building-only versus building-with-visible-road scenes

This would test whether mixed-scene composition predicts the observed directional failures without inventing new performance claims.

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Deployment architecture

Diagram options

The production artifact should contain or be paired immutably with the exact preprocessing, class order, calibration parameters, and decision-policy version.

Architecture table

Component	Deployment design	Verification or experiment
Input validation	Accept documented formats such as JPEG and PNG. Verify successful decoding, three-channel RGB conversion, finite pixel values, file-size and dimension limits, and protection against decompression bombs. Reject empty, corrupted, unsupported, or excessively large files with a clear error code.	Test malformed, grayscale, RGBA, CMYK, truncated, very small, and extremely large inputs before release.
Identical preprocessing	Convert to RGB, resize exactly to 150×150 using the same TensorFlow resize behavior, then apply <code>tf.keras.applications.mobilenet_v2.preprocess_input</code> . Do not apply random flip, rotation, zoom, or contrast during inference.	Store reference images and expected tensors/predictions; verify parity between notebook and serving pipeline.
Model artifact	Explicitly reload the chosen saved checkpoint and export a clean inference model. Confirm that it reproduces the reported predictions before deployment.	New experiment required: compare the saved best checkpoint with the final in-memory model because equivalence was not established in the existing artifacts.
Model versioning	Assign an immutable version containing model checksum, code version, TensorFlow version, class mapping, preprocessing version, calibration version, training-data fingerprint, and decision-policy version.	Require prediction-regression tests before registering a version.
Output contract	Return model version, predicted class, calibrated probability, review/abstention status, and reason code. Keep the class order fixed: buildings, forest, glacier, mountain, sea, street.	Validate the mapping against known reference cases.
Probability calibration	Fit temperature scaling or another simple calibration method using validation data only. Never fit calibration parameters on the test set.	New experiment required: compare uncalibrated and calibrated NLL, Brier score, ECE.

Calibration	data only. Never fit calibration parameters on the test set.	calibrated NLL, Brier score, ECE, and reliability diagrams.
Confidence thresholds	Use calibrated probabilities. Define separate thresholds for automatic acceptance, human review, and abstention. Do not infer thresholds from the six selected confident errors.	New experiment required: select thresholds from validation coverage-risk curves and operational error costs.
Unknown classes	Treat “none of the six classes” as an abstention problem; softmax always chooses one known class. Combine low confidence with an out-of-distribution detector. Return <code>unknown/review_required</code> , not a forced label.	New experiment required: build an OOD validation set and compare maximum calibrated probability, entropy, energy scores, and embedding-distance methods.
Human-review routing	Route low-confidence, OOD, invalid-but-recoverable, and policy-sensitive cases to review. Give priority to glacier–mountain and street–buildings disagreements. Reviewers may select one class, multiple plausible classes, unknown, or labeling error.	Pilot the queue to measure review volume, agreement, and turnaround time.
Logging	Record request/model versions, predicted probabilities, decision status, latency, validation failures, drift features, and eventual reviewer labels. Avoid retaining raw images unless authorized and necessary.	Perform privacy, retention, and access-control review before launch.
Serving reliability	Use bounded request sizes, timeouts, concurrency limits, health checks, warm model loading, and graceful failure. Separate inference failures from abstentions.	Load-test the complete serving path.
Rollout	Begin with shadow evaluation, then a small canary deployment, followed by gradual promotion. Keep the previous model immediately deployable.	Promotion requires passing quality, calibration, latency, and error-budget gates.

Monitoring table

Monitoring area	Metrics	Action or trigger
Overall quality	Accuracy, macro-F1, weighted-F1, loss, reviewed error rate	Investigate statistically credible or operationally significant degradation relative to the approved baseline. Numerical limits require production baselining.
Class-specific quality	Per-class precision, recall, F1, support, and prediction frequency	Alert when a class deteriorates even if aggregate accuracy remains stable.
Critical confusions	Mountain→glacier, glacier→mountain, street→buildings, buildings→street, glacier→sea	Track directional counts and rates separately. Escalate sustained increases after controlling for changes in class prevalence.
Calibration	NLL, Brier score, ECE, reliability by probability bin and class	Recalibrate or retrain when confidence no longer corresponds to observed correctness. New experiment required to establish initial baselines.
Abstention behavior	Automatic-acceptance rate, human-review rate, OOD rate, unresolved rate	Investigate sudden increases, queue overload, or an unsafe decrease in abstention accompanied by more errors.
Input drift	Resolution, aspect ratio, brightness, contrast, blur, compression, color statistics, file format, embedding distribution	Compare against training and approved-production reference distributions. Investigate sustained drift before retraining.

Concept drift	Performance within stable input subgroups; class-conditional error and calibration changes	Suspect concept drift when image statistics remain similar but the relationship between inputs and verified labels changes.
Unknown content	Reviewer-confirmed unknown categories, OOD-score distribution, forced-label attempts	Expand rejection coverage or consider taxonomy changes when recurring unknown categories become operationally important.
Human review	Queue size, age, reviewer agreement, override rate, class-specific overrides	Audit high disagreement and repeated overrides; these may indicate label ambiguity or an obsolete taxonomy.
Latency	Median, p95 and p99 end-to-end latency; preprocessing and inference time separately	Roll back or scale when service objectives are breached persistently. New experiment required to set hardware-specific objectives.
Resources	CPU/GPU utilization, memory, batch size, throughput, cold-start time, error rate	Benchmark CPU and GPU serving; consider quantization only after measuring its effect on class-level accuracy and calibration.
Data quality	Decode failures, unsupported formats, extreme sizes, duplicate requests	Investigate upstream pipeline changes or abuse patterns.
Feedback quality	Percentage of predictions receiving verified labels, time to label, reviewer confidence	Do not retrain from unverified user clicks or implicit feedback without validation.

Retraining triggers

Retraining should be considered when one or more of these conditions persist over a sufficiently large verified sample:

- Class-specific recall or F1 declines beyond an approved tolerance.
- Glacier–mountain or street–buildings confusion increases materially.
- Calibration error rises enough to invalidate confidence thresholds.
- OOD or human-review volume exceeds operational capacity.
- A recurring unknown category warrants taxonomy expansion.
- Input or concept drift is confirmed.
- Sufficient high-quality reviewed examples accumulate to support an update.

Exact numerical triggers require new experiments and production baseline data. Retraining should not be triggered by a small number of unverified complaints.

Rollback criteria

Roll back to the last approved version if a new release causes:

- A verified regression in overall or class-specific quality
- Worse glacier–mountain or street–buildings confusion
- Calibration failure that makes automatic acceptance unsafe
- Unexpected class-order or preprocessing mismatch
- Prediction differences from reference regression tests
- Excessive latency, memory use, crashes, or timeouts
- A large unexplained change in abstention or class distribution
- Failure of logging, auditability, or human-review routing

Rollback should restore the model, preprocessing, calibration, and threshold policy together; these components form one deployable version.

CHATGPT

1. Concise GitHub project summary

This project compares a custom convolutional neural network with a fine-tuned MobileNetV2 for classifying Intel scene images into six categories: buildings, forest, glacier, mountain, sea, and street.

The custom CNN achieved 80.10% test accuracy with a loss of 0.5698. Fine-tuned MobileNetV2 achieved 90.27% test accuracy with a loss of 0.2487 and a macro-F1 score of 0.9048. Model-failure analysis found that glacier–mountain and street–buildings were the most important confusion patterns. Grad-CAM was used to examine spatial sensitivity in selected high-confidence errors, while recognizing that these visualizations do not establish causal or human-like reasoning.

The repository includes the executed TensorFlow notebook, evaluation tables, confusion matrices, Grad-CAM visualizations, and machine-readable result summaries. It also documents reproducibility limitations, artifact inconsistencies, calibration requirements, and a deployment-monitoring design.

2. Publication-ready Medium article outline

Proposed title

Teaching a Neural Network to Recognize Natural Scenes: From a Custom CNN to Explainable Transfer Learning

Subtitle

A reproducible comparison of a custom TensorFlow CNN and fine-tuned MobileNetV2 on the Intel Image Classification dataset —with failure analysis, Grad-CAM, and lessons from iterative prompt engineering.

1. Introduction: Why scene classification is harder than it looks

- Introduce the six classes: buildings, forest, glacier, mountain, sea, and street.
- Explain that several labels can coexist visually within one image.
- State the project objective: compare a custom CNN with an ImageNet-initialized MobileNetV2.
- Clarify that the analysis focuses not only on accuracy but also on where and how the final model fails.

2. Framing the project with CRISP-DM

Organize the workflow around:

1. Business Understanding
2. Data Understanding
3. Data Preparation
4. Modeling
5. Evaluation
6. Deployment

Explain that the original notebook primarily addresses the technical classification problem; a real deployment would additionally require operational objectives, error costs, calibration, and monitoring.

3. Understanding the dataset

- 14,034 labeled images in the original training directory
- 11,228 images used for training
- 2,806 images used for validation
- 3,000 images used for testing
- Six classes with limited but nonzero class imbalance
- One representative image per class and a class-distribution visualization

Discuss limitations:

- No exact or perceptual duplicate audit
- No file-level split manifest
- No systematic label-ambiguity review
- No external-distribution evaluation

4. Preparing the images

- Resize images to 150×150
- Use a batch size of 64
- Apply horizontal flipping, rotation, zoom, and contrast augmentation during training
- Rescale custom-CNN inputs to [0, 1]
- Use `mobilenet_v2.preprocess_input` for MobileNetV2
- Keep the test dataset ordered with `shuffle=False`

Clarify that identical serving-time preprocessing is essential for deployment.

5. Model 1: Building a custom CNN

Describe:

- Four convolutional blocks
- Batch Normalization and max pooling
- Global average pooling
- A 128-unit dense layer
- 40% dropout
- 424,006 total parameters
- Adam optimization, early stopping, learning-rate reduction, and checkpointing

Report the verified test result:

- Accuracy: 80.10%
-

• Loss: 0.5698

Avoid diagnosing overfitting solely from non-epoch-aligned maximum training and validation accuracies.

6. Model 2: Transfer learning with MobileNetV2

Describe:

- ImageNet initialization
- Initially frozen convolutional base
- Global average pooling and 35% dropout
- Partial fine-tuning of the final portion of MobileNetV2
- Frozen Batch Normalization layers
- Fine-tuning learning rate of 10^{-5}

Explain that this staged design is intended to preserve pretrained representations while adapting higher-level features. State clearly that its benefit was not isolated through ablation.

7. Verified model comparison

Model	Test accuracy	Test loss	Macro-F1
Custom CNN	80.10%	0.5698	Not computed
Fine-tuned MobileNetV2	90.27%	0.2487	0.9048

Explain that MobileNetV2 achieved higher accuracy and lower loss in this run.

Do not claim that model complexity alone caused the improvement. Architecture, ImageNet initialization, preprocessing, optimization, parameter count, and training schedule all changed simultaneously.

8. Looking beyond overall accuracy

Present the MobileNetV2 class ranking by F1:

1. Forest: 0.9895
2. Sea: 0.9323
3. Street: 0.9169
4. Buildings: 0.9161
5. Mountain: 0.8424
6. Glacier: 0.8315

Clarify that street only narrowly exceeds buildings and that no uncertainty estimates are available.

9. Where the model fails

Present the largest directional confusions:

- Mountain → glacier: 69 images, 13.1%
- Glacier → mountain: 63 images, 11.4%
- Street → buildings: 49 images, 9.8%
- Glacier → sea: 30 images, 5.4%
- Buildings → street: 20 images, 4.6%

Discuss plausible visual explanations:

- Snow, ice, rock, ridgelines, and terrain texture overlap between glacier and mountain.

- Roads and buildings naturally coexist in urban images.
- Water and horizon cues may influence glacier → sea errors.

Label these explanations as hypotheses rather than established causes.

10. Model error or label ambiguity?

Explain the difference between:

- An evaluation error: prediction differs from the folder label
- An unequivocal semantic error: human reviewers agree that the prediction is visually wrong

Use mixed urban and mountain/glacier scenes to show why single-label classification can oversimplify the image.

Recommend blinded multi-reviewer relabeling and multilabel annotation as future validation.

11. What the confident errors reveal

Discuss the six displayed errors with confidence between 98.5% and 99.8%.

Correct interpretation:

- Some wrong predictions receive very high softmax scores.
- High softmax probability should not automatically be interpreted as a trustworthy correctness probability.

Important limitation:

- The examples were intentionally selected as confident mistakes.
- They do not prove that the entire model is globally miscalibrated.

12. What Grad-CAM shows—and what it does not

Grad-CAM supports a limited claim: the predicted-class score is sensitive to feature-map activations associated with particular image regions.

It cannot prove:

- Causal feature use
- Human-like concept recognition
- Explanation faithfulness
- Explanation stability
- Global model behavior
- Whether the dataset label is semantically unambiguous

Recommend occlusion tests, attribution-method comparisons, and explanation sanity checks.

13. An artifact-consistency lesson

Explain that the confusion-matrix image and exported report support:

- Mountain → glacier: 69
- Glacier → mountain: 63
- Street → buildings: 49

The source of one notebook cell instead contains 72, 52, and 50. The cause cannot be reconstructed from the artifacts.

Use this as a reproducibility lesson: derived summaries should be calculated directly from saved predictions rather than manually hard-coded.

14. From notebook to deployment

Describe the proposed production flow:

- Validate and decode inputs
- Convert to RGB
- Resize to 150×150
- Apply identical MobileNetV2 preprocessing
- Produce probabilities
- Apply a validation-fitted calibration layer
- Accept, abstain, or route to human review
- Monitor class-specific performance and drift

Mark calibration thresholds, OOD detection, latency requirements, and monitoring limits as requiring new experiments.

15. How iterative prompt engineering improved the project

Explain the progression:

- Start with a domain-role prompt
- Ground the analysis in attached artifacts
- Impose nonfabrication constraints
- Decompose the audit through CRISP-DM
- Require explicit output structures
- Challenge the first audit through skeptical peer review
- Narrow subsequent prompts to failure analysis and deployment
- Refine claims from broad interpretations to evidence-calibrated conclusions

Emphasize that prompt engineering improved analytical discipline and documentation, not the already-measured model scores.

16. Conclusion

- MobileNetV2 achieved stronger verified test performance in this run.
- Glacier–mountain confusion remains the central failure mode.
- Confident errors motivate calibration and abstention.
- Grad-CAM is useful for hypothesis generation but insufficient as causal evidence.
- Reproducibility requires saved predictions, exact split manifests, checkpoint verification, environment capture, and repeated runs.

3. Reproducibility statement

Reproducibility Statement

The executed experiment used TensorFlow 2.20.0, two Tesla T4 GPUs, a random seed of 42, an image size of 150×150, and a batch size of 64. Python, NumPy, and TensorFlow random seeds were set in the notebook. The labeled dataset was divided into 11,228 training images and 2,806 validation images, while a separate ordered test directory containing 3,000 images was used for final evaluation.

The notebook records the model architectures, preprocessing functions, augmentation layers, optimizers, learning rates, callbacks, training outputs, and evaluation procedure. The exported artifacts include model-comparison results, a MobileNetV2 classification report, confusion matrices, Grad-CAM visualizations, and a JSON summary.

These details improve repeatability but do not guarantee exact reproduction. The artifacts do not include file-level split manifests, dataset checksums, a complete dependency lock file, CUDA or cuDNN versions, deterministic-operation settings, saved-model checksums, or results from repeated seeds. Multi-GPU execution, GPU kernels, data prefetching, and stochastic augmentation may introduce nondeterminism.

One notebook source cell also contains manually specified confusion counts that disagree with the exported confusion matrix and JSON summary. The counts supported by the matrix, normalized percentages, class supports, recall values, and overall accuracy were used in the final analysis. The cause of the discrepancy cannot be determined from the available artifacts.

A fully reproducible release should preserve exact split manifests, sample-level predictions, environment metadata, dataset and model checksums, deterministic settings, and repeated-run results. The selected MobileNetV2 checkpoint should also be explicitly reloaded and verified against stored reference predictions before publication or deployment.

4. AI-assistance disclosure

AI-Assistance Disclosure

Generative AI was used as a technical-assistance and review tool during this project. It supported the structured audit of the executed notebook and exported results, identification of artifact inconsistencies, interpretation of model-failure patterns, development of reproducibility and deployment recommendations, and refinement of the project documentation.

The AI was instructed to treat the executed notebook, CSV files, JSON summary, confusion matrix, and Grad-CAM figure as the only authoritative evidence. It was explicitly prohibited from inventing or estimating experimental results. Numerical claims included in the final documentation were checked against the supplied artifacts, and disagreements between artifacts were disclosed rather than silently resolved.

AI-generated interpretations—such as possible visual explanations for glacier–mountain or street–buildings confusion—are presented as hypotheses, not experimentally established causes. Proposed calibration, out-of-distribution detection, robustness testing, repeated-seed evaluation, and deployment thresholds require new experiments and are not reported as completed results.

The author remains responsible for verifying the analysis, interpreting the findings, maintaining academic integrity, and ensuring that all published claims accurately reflect the executed experiment.

5. How iterative prompt engineering improved the project

How Iterative Prompt Engineering Improved the Project

Prompt engineering was used as an iterative analytical process rather than as a single request for polished writing. Each prompt introduced a clearer role, stronger evidence boundary, or more precise output requirement. This progressively changed the AI's contribution from a general notebook summary into a skeptical, traceable review of model performance, failure modes, reproducibility, and deployment readiness.

Role prompting

The initial prompt assigned the AI several relevant roles: senior computer-vision scientist, TensorFlow engineer, peer reviewer, technical educator, and CRISP-DM specialist. This established the expected technical depth and encouraged the analysis to address both implementation and scientific validity.

Role prompting did not make the AI's conclusions automatically correct. Its value was to define the perspective from which the evidence should be examined.

Context grounding

The AI was grounded in the executed notebook, CSV files, JSON summary, confusion matrix, and Grad-CAM figure. These artifacts were declared the only authoritative evidence.

This prevented the analysis from substituting expected benchmark performance, remembered dataset statistics, or unexecuted assumptions for the project's actual results. Context grounding also made it possible to detect that one notebook source cell contained confusion counts that disagreed with the exported artifacts.

Constraint specification

The prompts imposed explicit constraints:

- Do not fabricate or estimate results.
- Identify disagreements between artifacts.
- Separate measured evidence, inference, and recommendation.
- Do not rewrite the Medium article before completing the audit.
- Do not introduce unverified metrics.
- Mark deployment recommendations that require new experiments.

These constraints reduced unsupported certainty and made the boundary between completed work and future work visible.

Staged decomposition

The work was divided into successive stages instead of requesting everything at once:

1. Conduct a CRISP-DM audit.
2. Critique that audit as a skeptical peer reviewer.
3. Correct unsupported or overly strong claims.
4. Focus specifically on model failures.
5. Design deployment and monitoring controls.
6. Convert the verified analysis into publication materials.

This staged decomposition allowed each response to become evidence for the next review step. It also made methodological weaknesses easier to detect than they would have been in a single broad writing prompt.

Output contracts

Each prompt specified the required structure of the response. Examples included:

- Six CRISP-DM stages
- A verified results table
- Lists of findings, limitations, and improvements
- A claim-audit table with fixed columns
- Explicit “Observed evidence,” “Interpretation,” and “Proposed validation experiment” labels
- A concise architecture and monitoring table

These output contracts made omissions visible and forced conclusions to be paired with their evidentiary basis.

Evidence requirements

evidence requirements

The AI was required to cite exact notebook cells, metrics, tables, and images. It was also required to resolve artifact disagreements explicitly.

This requirement exposed an important reproducibility issue: the confusion counts written in one notebook source cell did not match the confusion matrix and exported JSON. The final analysis therefore used the values that were internally consistent with the matrix, normalized percentages, class supports, recalls, and overall accuracy while preserving the discrepancy as an unresolved provenance problem.

Adversarial self-critique

After the first audit, the AI was asked to review its own work as a skeptical peer reviewer. It classified earlier statements as measured, inferred, or unsupported and identified claims that needed correction.

For example, the initial language around overfitting, class separability, checkpoint identity, and calibration was softened because the existing artifacts did not fully establish those conclusions. This adversarial step reduced the risk that plausible interpretations would be mistaken for experimental findings.

Iterative refinement

Later prompts narrowed the analysis to specific questions:

- Which classes were strongest and weakest?
- What were the largest directional confusions?
- Which visual explanations were plausible?
- Which cases might involve label ambiguity?
- What could Grad-CAM support?
- What experiments would test those explanations?
- What safeguards would be required for deployment?

This refinement converted broad observations into testable hypotheses. Glacier–mountain similarity, architectural dominance in street images, probability calibration, and out-of-distribution handling were presented as questions for further experimentation rather than settled facts.

Overall, iterative prompt engineering improved the rigor, traceability, and communication of the project. It did not change the already-executed model or generate new experimental results. Its contribution was to make the analysis more disciplined: claims were tied to evidence, uncertainty was made explicit, inconsistencies were preserved, and future recommendations were separated from verified findings.